INFO 4310 – HW3

Alexander Peek (ap2298), Varsha Iyer (vvi2), Christina Zeng (cwz6)

Final Project Milestone 2

Introduction

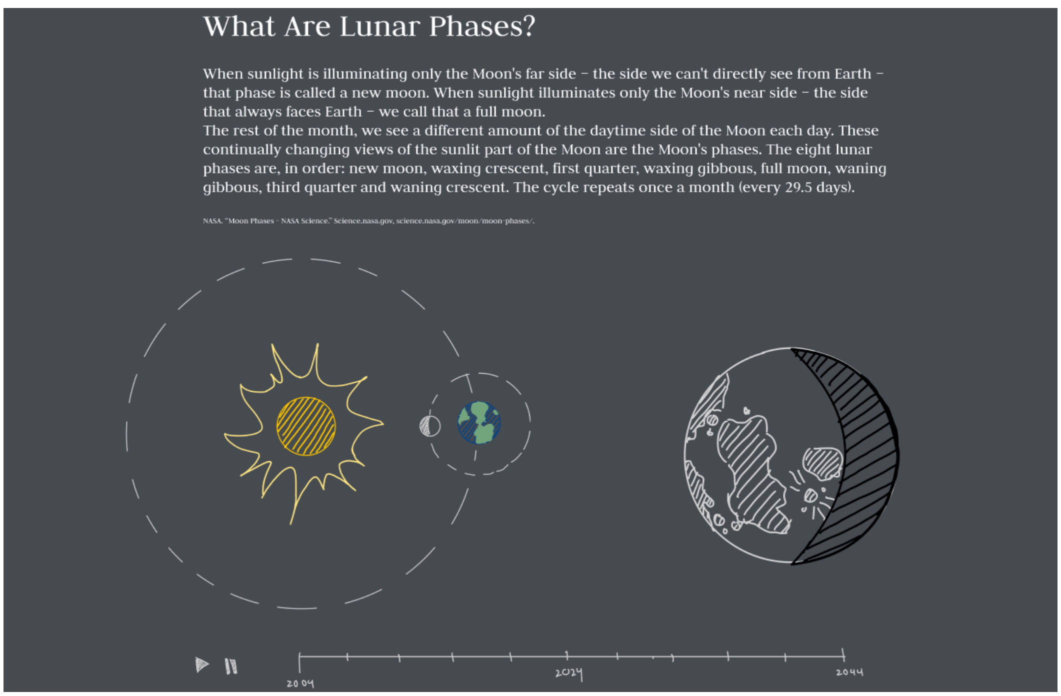
Our project focuses on the moon phase cycle and special moon events observed and predicted throughout the years 2004 - 2044 in Ithaca, NY as well as the effects the lunar cycle has on things such as tides, star visibility, and plant growth. Our target audience includes lunar enthusiasts who are interested in analyzing moon phase patterns in Ithaca, NY and its effects on Earthly matters. The intended goal is to convey an analysis of how the moon cycle has changed over time, as well as how we can adjust our practices involving tides, star visibility, and plant growth to be more aligned with the moon.

Dataset

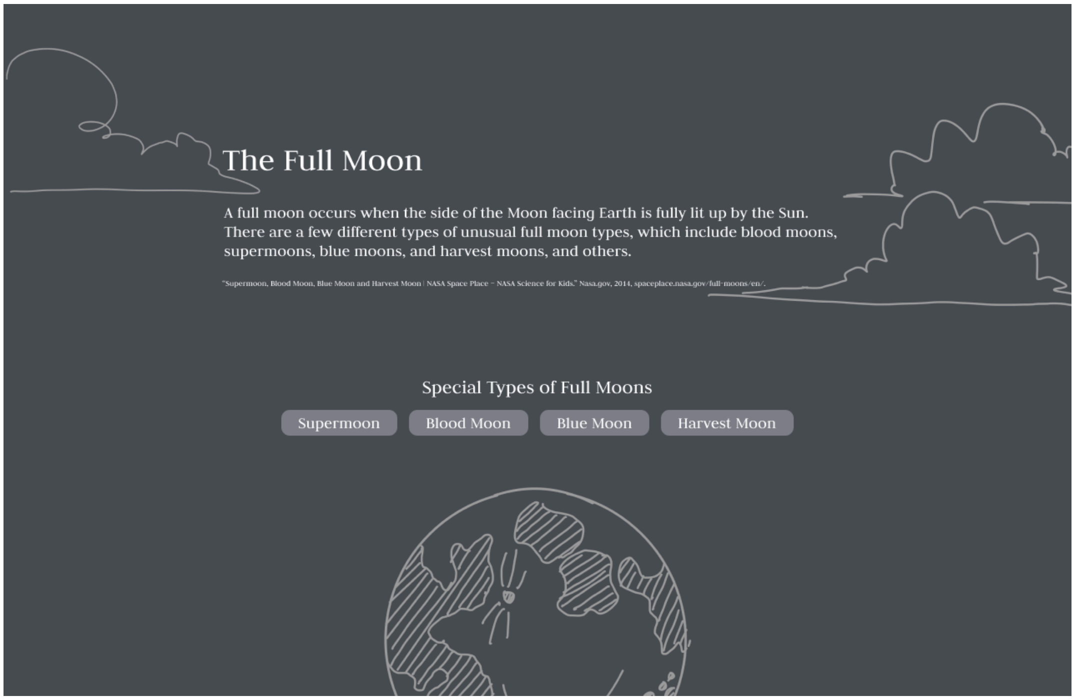
For our dataset, we scraped timeanddate.com to retrieve information about the exact dates of moon phases observed and predicted from 2004 to 2044 in Ithaca, NY. This website also lists various special moon events each year which we will also highlight in our visualizations. With this information, we manually built a json to use as our dataset in our project.

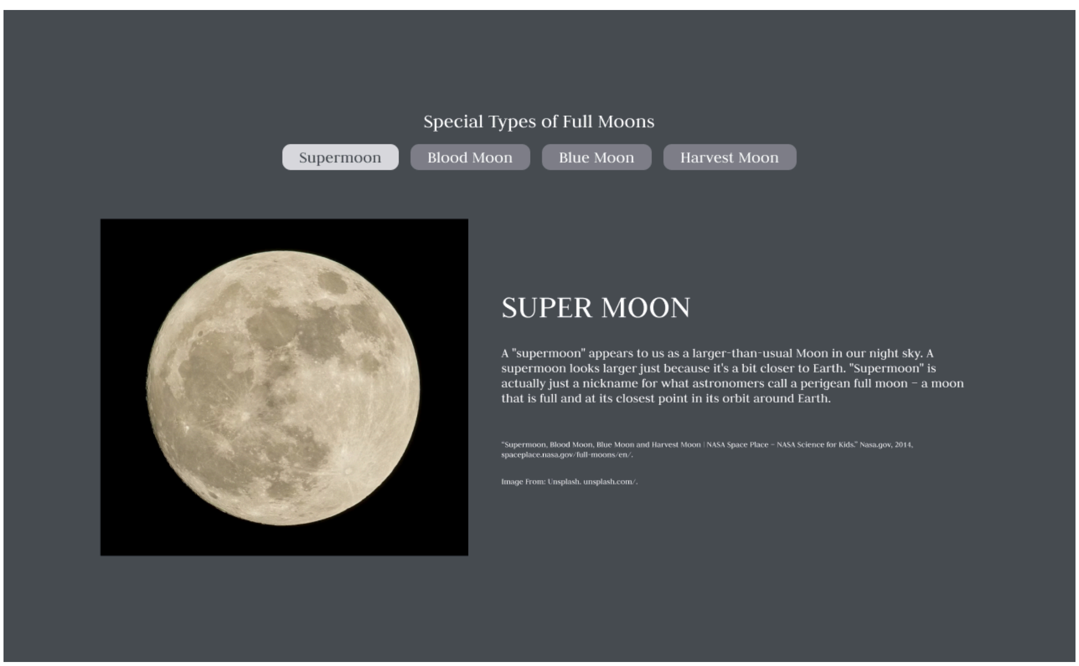
We currently have two versions of our dataset: moon_data.json and reverse_moon_data.json. Moon_data.json is formatted as { year: { "New Moon" : [jan date, feb date, mar date … dec date], "First Quarter" : [jan date, feb date, mar date … dec date] … "Third Quarter": …}, …}. Reverse_moon_data.json is formatted as { year: {jan date 1: moon phase 1, jan date 2: moon phase 2, jan date 3: moon phase 3, jan date 4: moon phase 4 … dec date 4: moon phase 4}, year 2: {...}, … }. Depending on how we decide to align our time slider and moon animation, we will use each version of the dataset accordingly.

Our Final Design



Seen in the two figures above, we took an educational approach with our project and explain how the moon works. This is where the moon phases come in when we explain the lunar cycle. Our visualization includes an animation where the moon is constantly rotating around the static Earth and the large moon figure will change to the corresponding moon phase depending on where in the rotation the moon is in the animation. We have a play and pause button so the user can start and stop the animation. We also hope to implement interaction with the time slider so users can know where the moon is in a specific year.





As the user scrolls, we will dive deeper into special events, specifically special types of full moons. The user can select each button to read more information about the selected full moon type. The button press will expand to display the information. Our second interaction for this visualization will be a zoom feature on the image of the special full moon.